

Topic 1[inheritance: use of super keyword]

Problem Statement: For the following Java program write

- i) to display x of class A in class B
- ii) statement to call getX() of class A in class B
- iii) to call parameterized constructor of in class B

Code	Output
<pre>class A { int x; public A() { x=0; } public A(int x) { this.x=x; } public int getX() { return (x+10); } } class B extends A { int x=20; B() { } B(int a) { super(a); } public int getX() { return (x+10); } int Display() { return super.x; } void getx_of_A() { System.out.println(super.getX()); } } public class inheritance { public static void main(String[] args) { B b =new B(15); System.out.println(b.Display()); b.getx_of_A(); } }</pre>	<pre>15 25 Press any key to continue . . .</pre>

Topic 2[interface in Java]

Problem Statement: Write three interfaces in Java namely AI, BI and CI with the following activities:

- i) Declare method PrintA() in AI
- ii) Declare method PrintB() in BI
- iii) Declare method PrintC() in CI
- iv) write body of PrintA() in class A
- v) write body of PrintB() in class B which is inherited from class A
- vi) write body of PrintC() in class C which is inherited from class B
- vii) Create object of class C in main() of First class and invoke PrintA(), PrintB() and PrintC()

Code	Output
<pre>interface AI { void PrintA(); } interface BI { void PrintB(); } interface CI { void PrintC(); } class A implements AI { @Override public void PrintA() { System.out.println("Method PrintA() in class A"); } } class B extends A implements BI { @Override public void PrintB() { System.out.println("Method PrintB() in class B"); } } class C extends B implements CI { @Override public void PrintC() { System.out.println("Method PrintC() in class C"); } } public class face { public static void main(String[] args) { C c = new C(); c.PrintA(); } }</pre>	Method PrintA() in class A Method PrintB() in class B Method PrintC() in class C

<pre> c.PrintB(); c.PrintC(); } }</pre>	
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Topic 3[java swing: Login Form]

Problem Statement: Write a Java program to produce a login form using Java swing features. The form checks login and password when click the submit button. If the input login and password are correct and it displays “Login Successful” message otherwise it displays “Invalid login/password” message.

Code	Output
<pre> import javax.swing.*; import java.awt.*; import java.awt.event.ActionEvent; import java.awt.event.ActionListener; public class login extends JFrame implements ActionListener { private JTextField usernameField; private JPasswordField passwordField; private JButton submitButton, refreshButton; public login() { setTitle("Login Form"); JPanel mainPanel = new JPanel(); mainPanel.setLayout(new GridLayout(3, 2)); JLabel usernameLabel = new JLabel(" Login "); usernameField = new JTextField(); JLabel passwordLabel = new JLabel(" Password "); passwordField = new JPasswordField(); submitButton = new JButton("Submit"); submitButton.addActionListener(this); refreshButton = new JButton("Refresh"); refreshButton.addActionListener(new ActionListener() { @Override public void actionPerformed(ActionEvent e) { usernameField.setText(""); passwordField.setText(""); } }) } }</pre>	

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    });
    mainPanel.add(usernameLabel);
    mainPanel.add(usernameField);
    mainPanel.add(passwordLabel);
    mainPanel.add(passwordField);
    mainPanel.add(submitButton);
    mainPanel.add(refreshButton);
    add(mainPanel);
    setSize(300, 150);
    setLocationRelativeTo(null);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
E);
}
@Override
public void actionPerformed(ActionEvent e)
{
    String username = usernameField.getText();
    String password = new
String(passwordField.getPassword());

    String correctUsername = "admin";
    String correctPassword = "password";

    if (username.equals(correctUsername) &&
password.equals(correctPassword))
    {
        JOptionPane.showMessageDialog(this, "Login
Successful", "Success",
JOptionPane.INFORMATION_MESSAGE);
    }
    else
    {
        JOptionPane.showMessageDialog(this, "Invalid
login/password", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

public static void main(String[] args)
{
    SwingUtilities.invokeLater(() ->
    {
        login loginForm = new login();
        loginForm.setVisible(true);
    });
}
}

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